

IS883: Deploying Generative AI

Mohannad Elhamod

Neural Nets in Language Modeling

Continued...

Fast Forward...

As neural networks arrived at the scene, they were utilized for language modeling.

- N-grams look for exact prefixes, which is limiting...
- However, neural networks can learn more interesting relationships between the words.

Example: All humans need water.

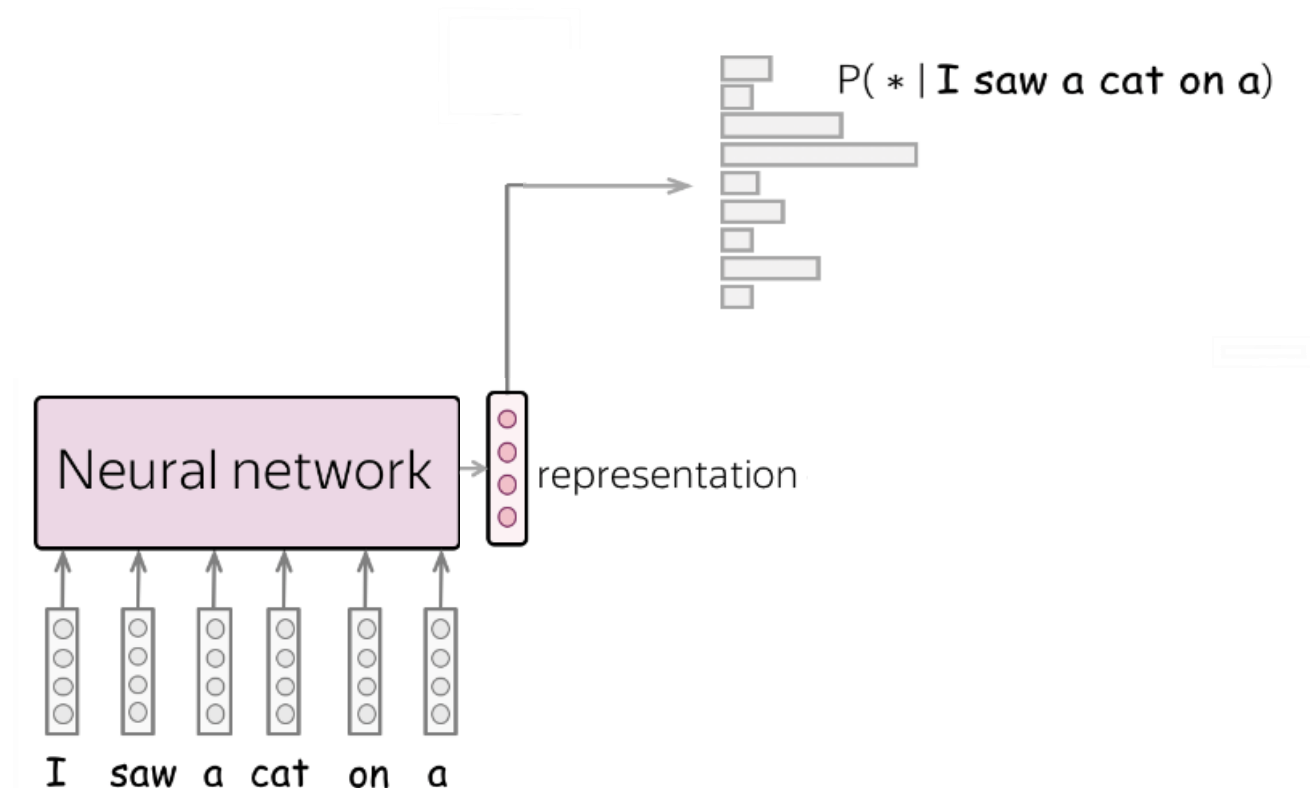
Mohannad is a human.

Therefore, Mannad needs water

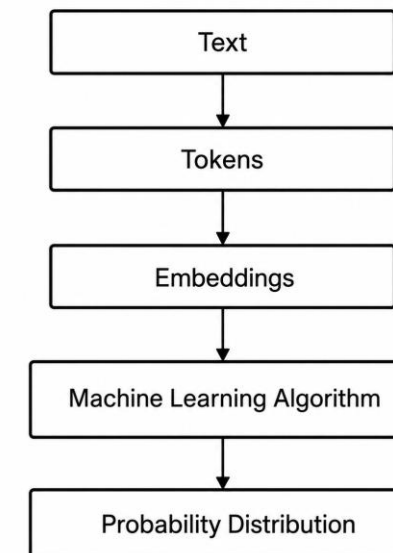
How to Represent Text?

- Machine Learning algorithms do not understand plain text...
- Hence, text needs to be represented in a *numeric* format that *retains the necessary information* in it.
- We call such a representation *an embedding*.
- To obtain text embeddings, let's take two steps:
 - We break the text into pieces called *tokens*...
 - This is called Tokenization. ([Demo](#))
 - Each token is represented as an embedding...
 - This is called Vectorization.

General Model Architecture



[Lena-volta](#)

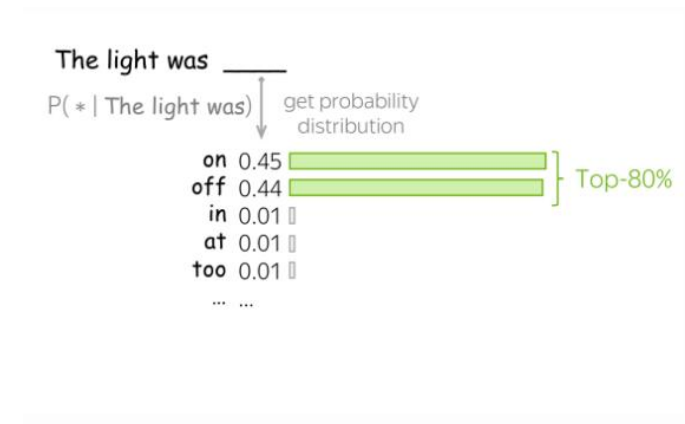


Word Embeddings

- We ideally want related words (i.e., *similar* meanings) to have smaller distances.
- [Demo](#)

Sampling The Distribution

- Always take top probability?
 - That makes the model deterministic (no creativity).
- Alternative?
 - Top-k, top-p, or [beam search](#)



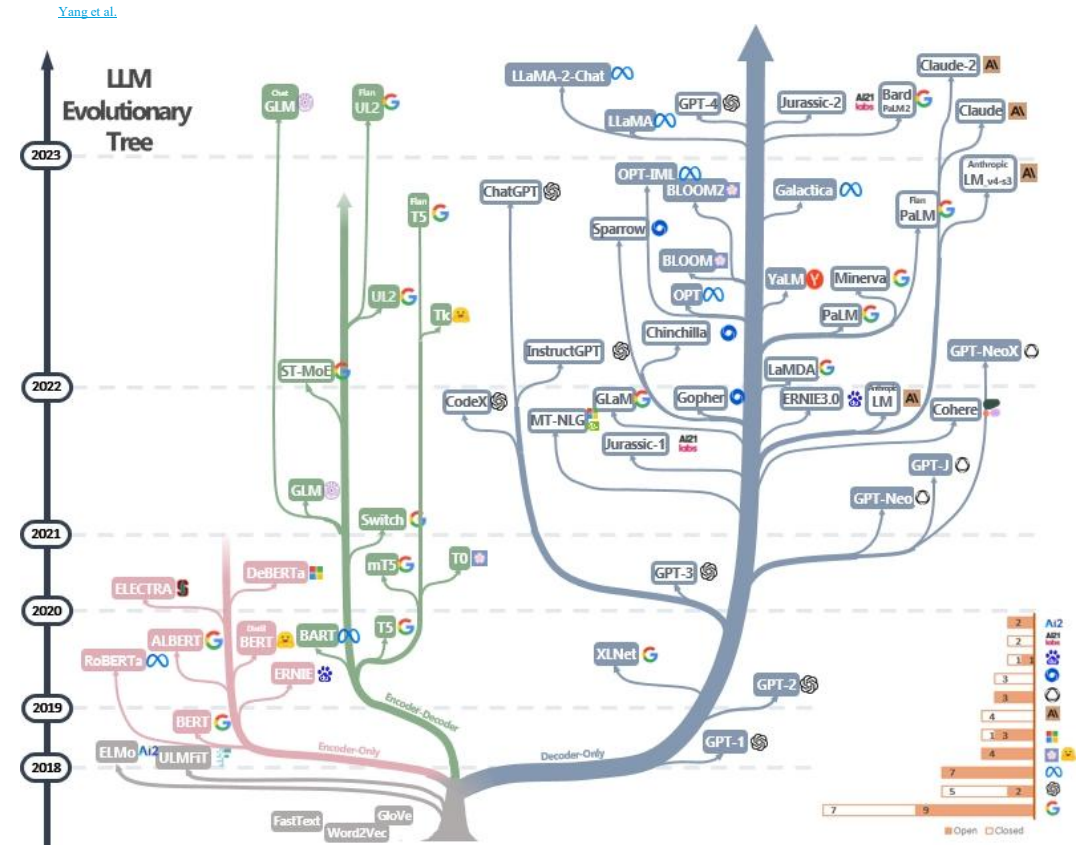
Measuring The Metric

- What are we looking for?
 - A model that is not surprised by the text it is seeing.
- We use perplexity.
 - Takes values between 1 and the number of possible tokens.
 - Smaller is better.
 - [Perplexity calculations: Demo](#)
 - [Next word probability: Demo](#)

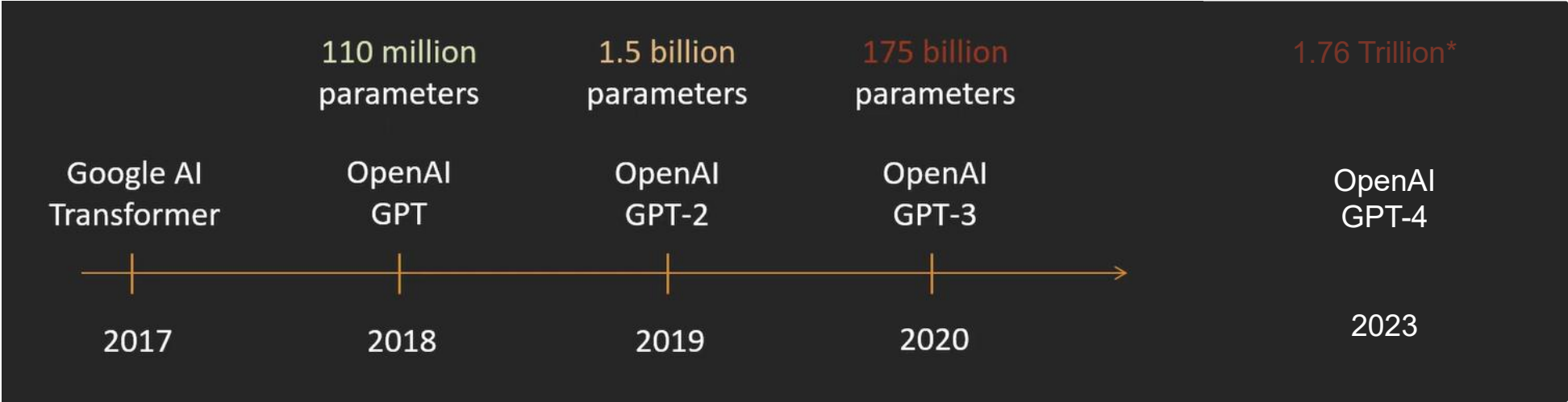
Models in the wild

Large Language Models

- LLMs are deep learning models with a special kind of architecture.
- **Data**.
- **Model** type and size.
- **Hyperparameters** (context size, embedding size,...).
- **Training process** (the cost function, fine-tuning, human feedback, etc.).



These Models Are LARGE!!!



[AI Coffee Break with Letitia](#)

[Book Corpus](#)
[WebText](#)

1,038 books (around 74M sentences and 1G words) of 16 different sub-genres (e.g., Romance, Historical, Adventure, etc.)

[Common Crawl + ...](#)

Over 240 billion pages.
Petabytes of data.

[????](#)

** Reported — never officially confirmed*



These Models Are EXPENSIVE!!!

\$78.4M

GPT-4 (2023)

\$191.4M

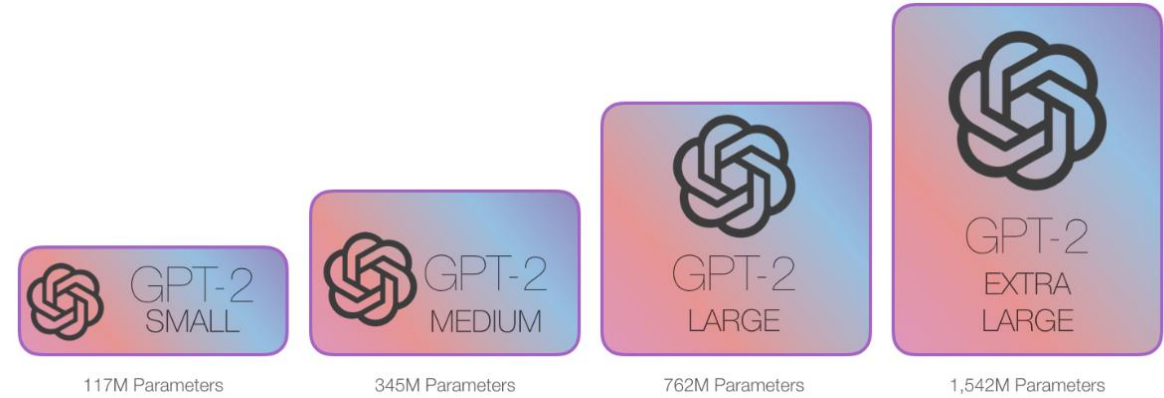
Gemini Ultra (2023)

estimated cost of the compute used to train

These are estimates, not disclosed figures. Stanford researchers worked them out from reported hardware and cloud rental prices — no company has published what it actually paid.

Stanford HAI, AI Index Report 2024, Chapter 1

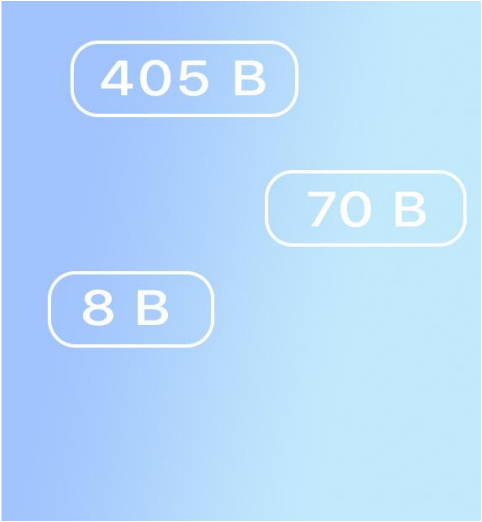
Different Model Sizes



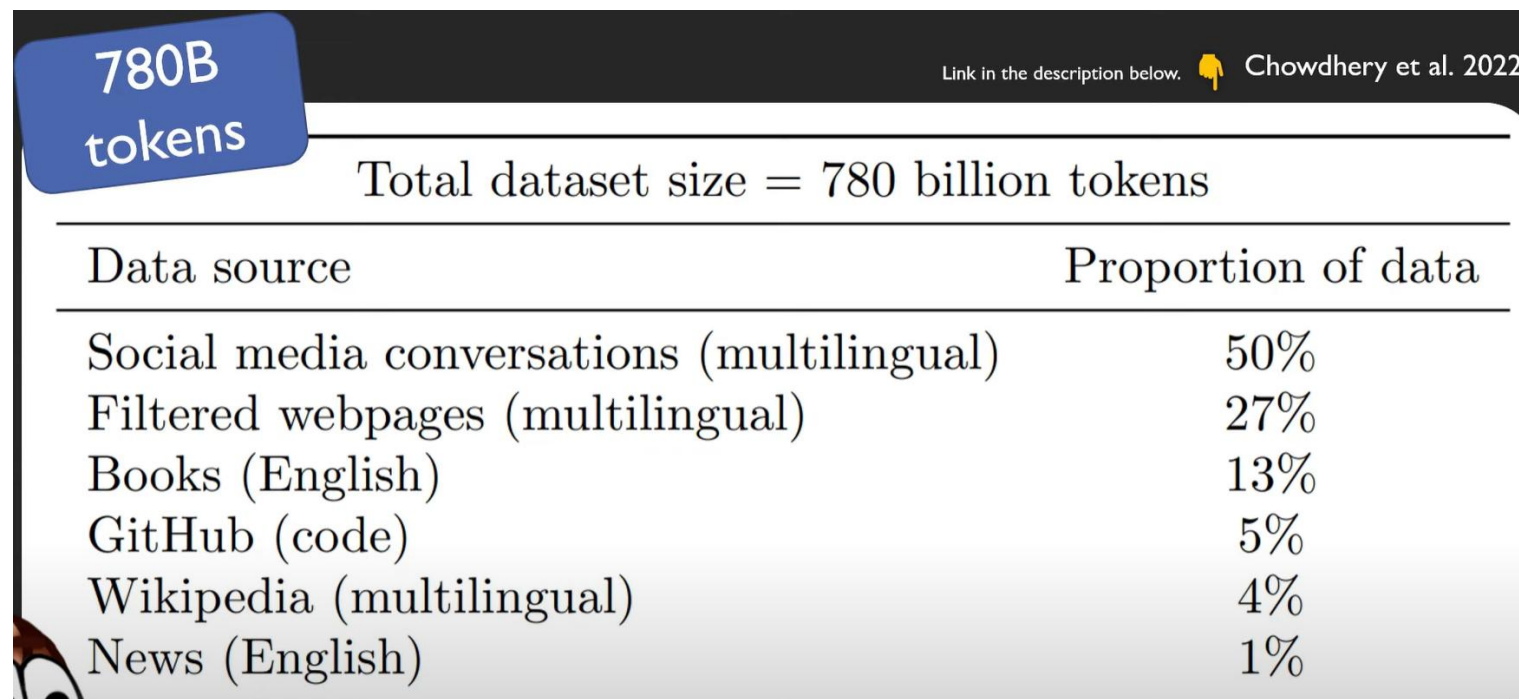
[Jay Alamar](#)

 Meta
Llama 3.1

Enter your prompt



Data is Large and Diverse...



780B tokens

Link in the description below. 📌 Chowdhery et al. 2022

Total dataset size = 780 billion tokens

Data source	Proportion of data
Social media conversations (multilingual)	50%
Filtered webpages (multilingual)	27%
Books (English)	13%
GitHub (code)	5%
Wikipedia (multilingual)	4%
News (English)	1%

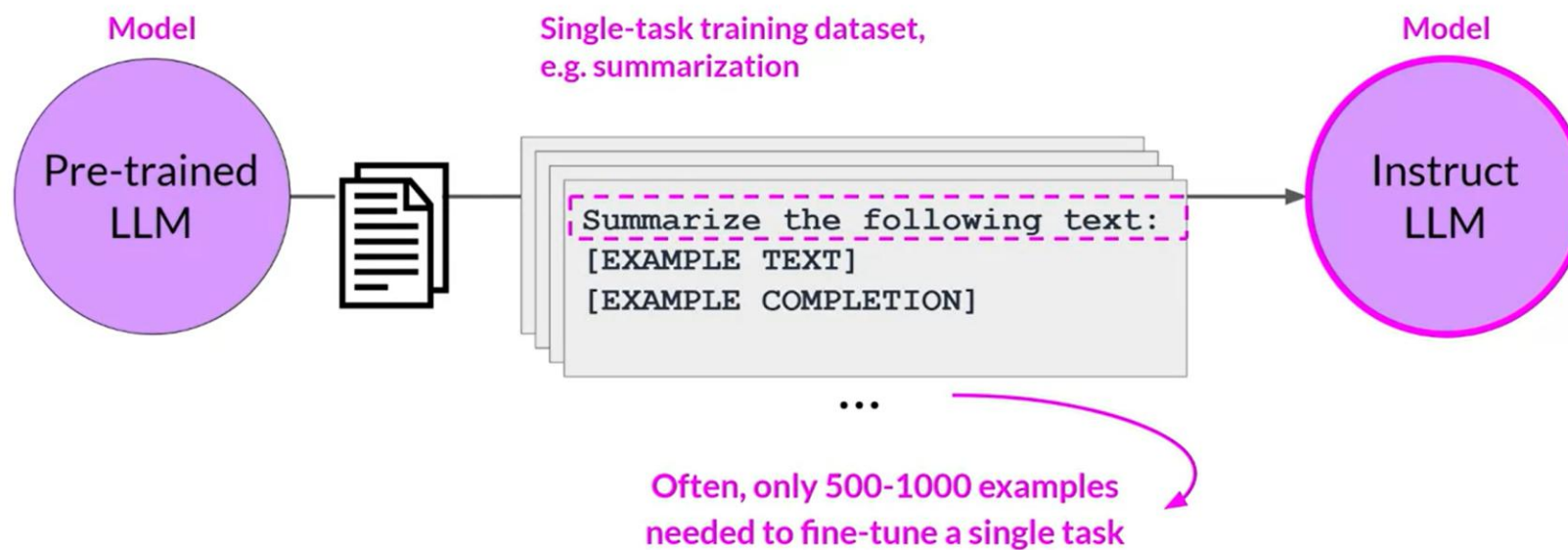
Google's PaLM pretraining dataset

[AI Coffee Break with Letitia](#)

Pre-trained Models: Democratizing AI

- Most of us don't have the expertise, data, or resources to train anything close to these impressive large models.
- Instead:
 - *Zero-shot Learning*: We can use others' models out-of-the-box, even though they have never seen our data before.
 - *Fine-Tuning*: Others' models be used as a base for further training (e.g., if the training data is non-public legal documents).

Example: Instruct LLMs



[Coursra](#)

In-Class Work

Run a model on your own laptop

```
ollama run llama3.2:3b
```

HuggingFace

What is it for?

- To download and use a pre-trained model locally.
- *To download and fine-tune a model with your own data.)*
- To use *some* models through an API.



Hugging Face

Exploring HuggingFace Land

- [HuggingFace tasks](#)
- [HuggingFace models](#)

In-Class Work

HuggingFace

Resources

- [Video: LLMs vs The Brain](#)
- [Video: Deciding which pre-trained model to fine-tune](#)